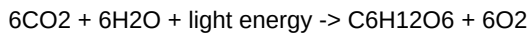


Photosynthesis: The Process of Life

Photosynthesis is the biological process by which green plants, algae, and certain bacteria convert light energy into chemical energy stored in glucose. This process is fundamental to life on Earth.

The Chemical Equation:



Key Components:

1. Chlorophyll: The green pigment in chloroplasts that absorbs light energy.
2. Light Reactions: Occur in the thylakoid membranes, producing ATP and NADPH.
3. Calvin Cycle: Occurs in the stroma, using ATP and NADPH to fix CO₂ into glucose.

Light-Dependent Reactions:

- Water molecules are split (photolysis), releasing O₂
- Light energy excites electrons in photosystem II and photosystem I
- Electron transport chain generates ATP via chemiosmosis
- NADP⁺ is reduced to NADPH

Calvin Cycle (Light-Independent Reactions):

- Carbon fixation: CO₂ is attached to RuBP by the enzyme RuBisCO
- Reduction: 3-PGA is reduced to G3P using ATP and NADPH
- Regeneration: RuBP is regenerated to continue the cycle
- For every 3 CO₂ molecules fixed, 1 G3P molecule exits the cycle

Factors Affecting Photosynthesis:

- Light intensity: Rate increases with intensity up to a saturation point
- CO₂ concentration: Higher CO₂ generally increases rate
- Temperature: Optimal range is 25-35°C for most plants
- Water availability: Drought reduces stomatal opening, limiting CO₂ intake

Common Misconceptions:

1. Plants only photosynthesize during the day but respire all the time
2. Photosynthesis does NOT occur at night (requires light)
3. Oxygen is a byproduct, not the main product
4. The glucose produced is used for cellular respiration AND building biomass

Applications in Agriculture:

- Greenhouse farming optimizes light, CO₂, and temperature
- C₄ plants (corn, sugarcane) have evolved to minimize photorespiration
- CAM plants (cacti, pineapple) open stomata at night to conserve water